10.5.9

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Question

For a matrix

$$\mathbf{M} = \begin{pmatrix} \frac{4}{5} & \frac{3}{5} \\ \frac{3}{5} & x \end{pmatrix} \tag{0.1}$$

The transpose of the matrix is equal to inverse of the matrix, i.e.,

$$\mathbf{M}^\top = \mathbf{M}^{-1} \tag{0.2}$$

The value of x is given by

Solution

From (0.2) it is evident that the matrix (0.1) is an orthogonal matrix. We can proceed by the fact that columns of an orthogonal vector are perpendicular

$$\mathbf{M} = \begin{pmatrix} \mathbf{c_1} & \mathbf{c_2} \end{pmatrix} \tag{0.3}$$

where

$$\mathbf{c_1} = \begin{pmatrix} \frac{4}{5} \\ \frac{3}{5} \end{pmatrix}, \mathbf{c_2} = \begin{pmatrix} \frac{3}{5} \\ x \end{pmatrix} \tag{0.4}$$

Solution

By using (0.2):

$$
\begin{aligned}
\mathbf{c_1}^\top \mathbf{c_2} &= 0 && (0.5)\\
\implies \begin{pmatrix} \frac{4}{5} & \frac{3}{5} \end{pmatrix} \begin{pmatrix} \frac{3}{5} \\ x \end{pmatrix} &= 0 && (0.6)\\
\implies \frac{12}{25} + \frac{3x}{5} &= 0 && (0.7)\\
\implies x &= -\frac{4}{5} && (0.8)
\end{aligned}
$$